MICHAEL WARUIRU

Backend Developer | AI Engineer

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PROFESSIONAL SUMMARY

Self-taught **Backend Developer** with strong expertise in Golang and Python, recently expanding into **AI/ML** with TensorFlow and Scikit-learn. Experienced in building scalable applications, integrating APIs, and implementing ML-driven projects (e.g. Plant Disease Detector and document analyzer). Passionate about applying AI to real-world problems, eager to contribute as an AI Engineer (intern/entry-level) and building scalable backend architecture.

TECHNICAL SKILLS

- **AI/ML Tools & Frameworks**: TensorFlow, Scikit-learn, spaCy, Matplotlib, Pandas
- **Programming Languages**: Python, Golang
- **Frameworks**: Django, Flask
- **Databases**: MySQL
- **Cloud & Deployment**: Netlify, Render, Docker
- **Others**: REST APIs, Authentication, M-Pesa integration

PROJECTS

- **Plant Disease Detector** – Built a TensorFlow/Keras model for detecting potato, tomato, and pepper leaf diseases, outputting treatment recommendations.
- **Document analyzer** – Built with spaCy and Flask for detecting risky clauses in contracts, agreements, etc.
- **Sacco Management System** – Designed a Golang + MySQL system to manage vehicles, drivers, and checkpoints.
- **AI Chatbot** – Chatbot for getting real-time insights and analytics from the database.
- **Library Management System** – Automated catalog management, book borrowing/return using Go.

WORK EXPERIENCE

Lead Trainer | *Segera Ventures* | **Apr 2025 – Present**

- Trains government officials from all South Sudan states (including state directors, county officers, cooperative leaders and FAO officials) on the **Co-op Magic System**, an FAO-supported digital platform for managing cooperatives and agricultural data.
- Conducts **hands-on training sessions** covering system onboarding, data entry workflows, cooperative registration, loan record management, and reporting dashboards.
- Provides **on-site technical support** and acted as a **first-level system tester**, identifying bugs, documenting issues, and forwarding actionable feedback to the development team.
- Develops simplified **user guides** and training materials tailored to varying digital-literacy levels among state officials.
- Leads **capacity-building workshops** in remote field locations, ensuring that officers could operate the system independently after deployment.
- Coordinates with FAO project managers and Ministry of Agriculture staff to ensure smooth rollout and adoption across states.
- Assists in monitoring system usage, gathering user feedback, and recommending enhancements for future updates.
- Currently building an **AI Chatbot** using LangChain and LangGraph for the existing system to help in navigating through the system and getting real-time insights and analytics.

Backend Developer Intern | *Skevatech Ltd* | **Jul 2023 – Oct 2024**

- Developed full-stack web apps in Golang & Go templates like **Sacco management** and **library system** to be implemented to an existing school system.
- Integrated M-Pesa payment solutions.
- Conducted end-user training for HR system.
- Built APIs & authentication-based personal projects.

Digitization Team Member | *Ministry of Lands* | **Mar 2020 – Nov 2021**

- Participated in national land records digitization.
- Ensured accuracy and security of digital databases.

EDUCATION & CERTIFICATIONS

- **AI Engineering Certificate** – PLP Academy (May – July 2025)
- **Full-stack Software Development** – PLP Academy (Feb – May 2025)
- **Python Development** – Udemy (2023)
- J**avaScript, HTML & CSS** – Udemy (2023)
- **Bachelor of Economics & Finance** – Kenyatta University (May 2013 – Dec 2017)

SOFT SKILLS

• Communication • Adaptability • Time Management • Problem-solving

HOBBIES & INTERESTS

Backend Development | Machine Learning | Web Development | Cycling | Writing Blogs | Traveling

REFERENCES

Available upon request.